

Introductory Trigonometry Formula Sheet

Unit Circle, Right Triangles, Periodicity, and Identities

Unit Circle

At angle θ , the point is

$$(x, y) = (\cos \theta, \sin \theta), \quad \tan \theta = \frac{y}{x}.$$

Degrees	Radians	$(\cos \theta, \sin \theta)$
0°	0	$(1, 0)$
30°	$\frac{\pi}{6}$	$(\frac{\sqrt{3}}{2}, \frac{1}{2})$
45°	$\frac{\pi}{4}$	$(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2})$
60°	$\frac{\pi}{3}$	$(\frac{1}{2}, \frac{\sqrt{3}}{2})$
90°	$\frac{\pi}{2}$	$(0, 1)$
120°	$\frac{2\pi}{3}$	$(-\frac{1}{2}, \frac{\sqrt{3}}{2})$
135°	$\frac{3\pi}{4}$	$(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2})$
150°	$\frac{5\pi}{6}$	$(-\frac{\sqrt{3}}{2}, \frac{1}{2})$
180°	π	$(-1, 0)$
210°	$\frac{7\pi}{6}$	$(-\frac{\sqrt{3}}{2}, -\frac{1}{2})$
225°	$\frac{5\pi}{4}$	$(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2})$
240°	$\frac{4\pi}{3}$	$(-\frac{1}{2}, -\frac{\sqrt{3}}{2})$
270°	$\frac{3\pi}{2}$	$(0, -1)$
300°	$\frac{5\pi}{3}$	$(\frac{1}{2}, -\frac{\sqrt{3}}{2})$
315°	$\frac{7\pi}{4}$	$(\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2})$
330°	$\frac{11\pi}{6}$	$(\frac{\sqrt{3}}{2}, -\frac{1}{2})$
360°	2π	$(1, 0)$

Quadrant signs

QI: all positive; QII: sine; QIII: tangent; QIV: cosine.

Angle conversion

$$180^\circ = \pi, \quad \text{degrees} \cdot \frac{\pi}{180} = \text{radians}.$$

SOH-CAH-TOA

$$\sin \theta = \frac{\text{opp}}{\text{hyp}}, \quad \cos \theta = \frac{\text{adj}}{\text{hyp}}, \quad \tan \theta = \frac{\text{opp}}{\text{adj}}.$$

$$\csc \theta = \frac{1}{\sin \theta}, \quad \sec \theta = \frac{1}{\cos \theta}, \quad \cot \theta = \frac{1}{\tan \theta}.$$

Pythagorean theorem and identities

$$a^2 + b^2 = c^2$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$1 + \tan^2 \theta = \sec^2 \theta, \quad 1 + \cot^2 \theta = \csc^2 \theta.$$

Fundamental periods

Functions	Period
$\sin, \cos, \sec, \csc$	2π
$\tan, \cot$	π

If $f(x)$ has period P , then $f(bx)$ has period

$$\frac{P}{|b|}.$$

Thus $f(x + P) = f(x)$ whenever both sides are defined.

Complement (cofunction) rules

For θ in degrees, replace $\frac{\pi}{2} - \theta$ below by $90^\circ - \theta$.

$$\sin\left(\frac{\pi}{2} - \theta\right) = \cos \theta, \quad \cos\left(\frac{\pi}{2} - \theta\right) = \sin \theta$$

$$\tan\left(\frac{\pi}{2} - \theta\right) = \cot \theta, \quad \cot\left(\frac{\pi}{2} - \theta\right) = \tan \theta$$

$$\sec\left(\frac{\pi}{2} - \theta\right) = \csc \theta, \quad \csc\left(\frac{\pi}{2} - \theta\right) = \sec \theta.$$

Undefined values

A reciprocal function is undefined when its denominator is zero:

$$\tan, \sec : \cos \theta = 0, \quad \cot, \csc : \sin \theta = 0.$$